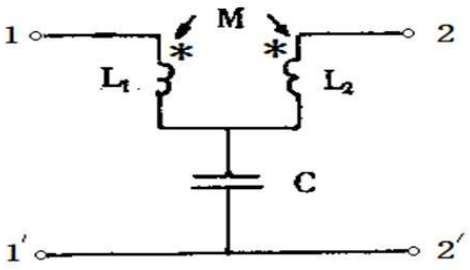
习题课6

- 二端口网络
- 第十一-十三章作业讲评

第6次习题课

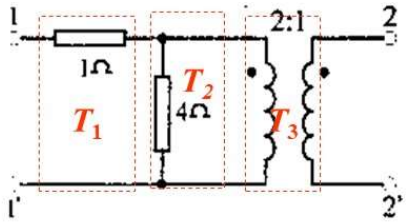
1　如图利用二端口联接构成的复合二端口，求图（a）的Z参数矩阵及图（b）的T参数矩阵。



$$Z' = \begin{bmatrix} j\omega L_1 & j\omega M \\ j\omega M & j\omega L_2 \end{bmatrix}$$

$$Z'' = \begin{bmatrix} \frac{1}{j\omega C} & \frac{1}{j\omega C} \\ \frac{1}{j\omega C} & \frac{1}{j\omega C} \end{bmatrix}$$

$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} Z'_{11} & Z'_{12} \\ Z'_{21} & Z'_{22} \end{bmatrix} + \begin{bmatrix} Z''_{11} & Z''_{12} \\ Z''_{21} & Z''_{22} \end{bmatrix}$$

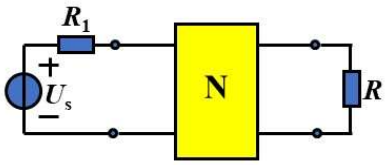


$$\mathbf{T}_1=\begin{bmatrix}1 & 1\ \Omega\\ 0 & 1\end{bmatrix}\quad \mathbf{T}_2=\begin{bmatrix}1 & 0\\ 0.25\ \mathrm{S} & 1\end{bmatrix}\quad \mathbf{T}_3=\begin{bmatrix}2 & 0\\ 0 & 1/2\end{bmatrix}$$

$$[\mathbf{T}]=[\mathbf{T}_1]\,[\mathbf{T}_2]\,[\mathbf{T}_3]=\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\ 0.25 & 1\end{bmatrix}\begin{bmatrix}2 & 0\\ 0 & 0.5\end{bmatrix}=\begin{bmatrix}2.5 & 0.5\ \Omega\\ 0.5\ \mathrm{S} & 0.5\end{bmatrix}$$

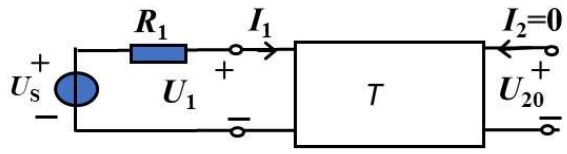
2. 如图所示，二端口网络N的传输参数为 $T=\begin{bmatrix}2 & 8\Omega\\ 0.5\,\mathrm{S} & 2.5\end{bmatrix}$

U_S=6V和 R_1=2Ω 的串联支路连接到端口 1, R 连接到端口 2。
求 R 获得最大功率时的值，并求此最大功率。



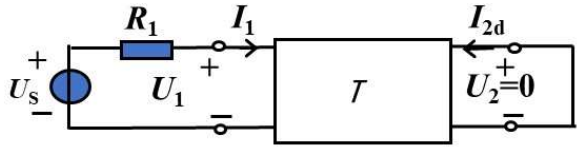
法 1:

$$T=\begin{bmatrix}2 & 8\Omega\\ 0.5\,\mathrm{S} & 2.5\end{bmatrix}$$



$$\begin{bmatrix}U_1=6-2I_1\\ I_1\end{bmatrix}=\begin{bmatrix}2 & 8\\ 0.5 & 2.5\end{bmatrix}\begin{bmatrix}U_{20}\\ 0\end{bmatrix}$$

开路电压: $U_{20}=2\mathrm{V}$

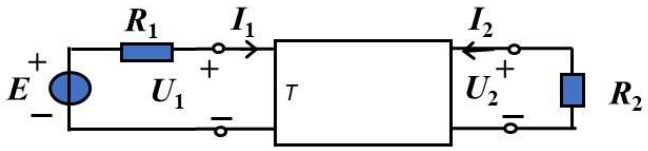


$$\begin{bmatrix}U_1=6-2I_1\\ I_1\end{bmatrix}=\begin{bmatrix}2 & 8\\ 0.5 & 2.5\end{bmatrix}\begin{bmatrix}0\\ -I_{2d}\end{bmatrix}$$

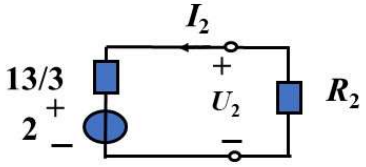
短路电流: $I_{2d}=-6/13\mathrm{A}$

$$R_{等}=\frac{U_{20}}{-I_{2d}}=\frac{13}{3}\Omega \qquad P_{\max}=\frac{U_{20}^2}{4R}=0.23\mathrm{W}$$

法 2:



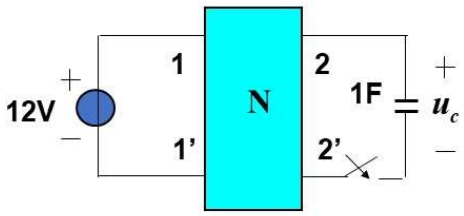
$$\begin{aligned}U_1&=2U_2-8I_2\\ I_1&=0.5U_2-2.5I_2\\ U_1&=6-2I_1\end{aligned}\quad\Rightarrow\quad U_2=\frac{13}{3}I_2+2$$



$$R=\frac{13}{3}\Omega \qquad P_{\max}=0.23\mathrm{W}$$

3. 二端口N的传输参数矩阵为 $Z=\begin{bmatrix}4 & 2\\ 2 & 5\end{bmatrix}\Omega$。

t=0时刻闭合开关，已知$u_c(0^-)$=1V，求$u_c(t)$。

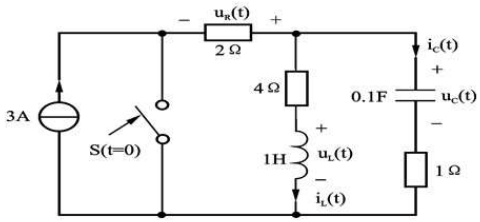


$$U_1 = 4I_1 + 2I_2$$
$$U_2 = 2I_1 + 5I_2 \Rightarrow U_2 = 6+4I_2$$
$$U_1 = 12$$

$$U_C(\infty) = 6\text{V}$$
$$\tau = R_{eq}C = 4s$$

$$u_C(t) = 6 - 5e^{-0.25t} \quad \text{V} \quad t \geq 0$$

11-1在图电路中，开关S原来打开，在 $t = 0$ 时合上，开关动作前电路已达稳态，试求当开关S闭合后一瞬间的 $u_C(0_+)$ $i_L(0_+)$ $u_R(0_+)$ $u_L(0_+)$ 和 $i_C(0_+)$



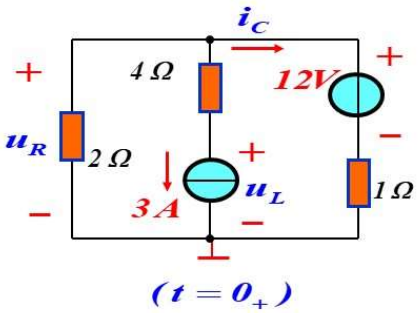
解

$t = 0_-$ 时： $i_L(0_-) = 3A$ $U_C(0_-) = 12V$

$t = 0_+$ 时： $i_L(0_+) = i_L(0_-) = 3A$

$U_C(0_+) = U_C(0_- = 12V$

0+瞬时电路如下

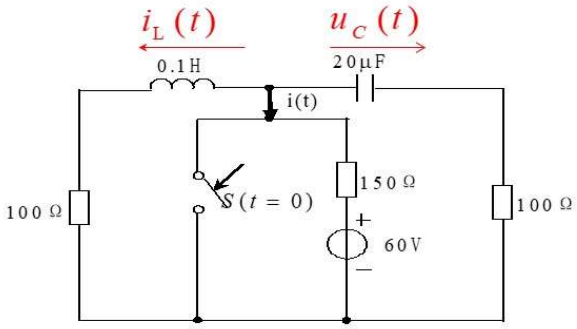


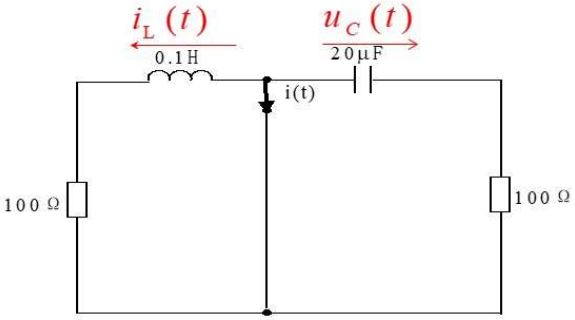
$$U_R(0_+) = \frac{\frac{U_C(0_+)}{1} - i_L(0_+)}{\frac{1}{2}+1} = \frac{12\times 2 - 3\times 2}{3} = 6V$$

$$U_L(0_+) = U_R(0_+) - i_L(0_+)\times 4 = 6 - 12 = -6V$$

$$i_C(0_+) = \frac{U_R(0_+) - 12}{1} = -6A$$

11-2 图所示电路开关S在t=0时闭合，开关动作之前电路已处于稳定状态，试求 t 时0的电流 $i(t)$





$i_L(0_-) = 0.24\text{A}$ $\tau_L = 0.001s$ $i_L(t) = 0.24e^{-1000t}A$

$u_C(0_-) = 60 - i_L \times 150 = 24\text{V}$ $\tau_C = 0.02s$ $u_C(t) = 24e^{-500t}\text{V}$

$$i(t) = -C\frac{du_C(t)}{dt} - i_L(t) = 0.24e^{-500t} - 0.24e^{-1000t}A$$

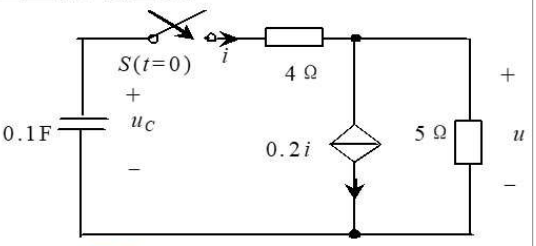
11-3 电路如图所示，电路原来已达稳态，$u_C(0_-) = 4V$

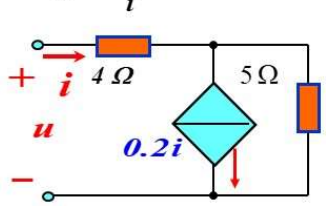
t=0时开关闭合。求t≥0时的电容电压u_C和5Ω电阻上的电压u。

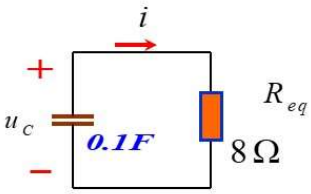
解 **(1) 求时间常数 τ**

$u = 4i + 5(i - 0.2i) = 8i$

$R_{eq} = \frac{u}{i} = 8\Omega$ $\tau = RC = 0.8S$







R_{eq}

0+后电路

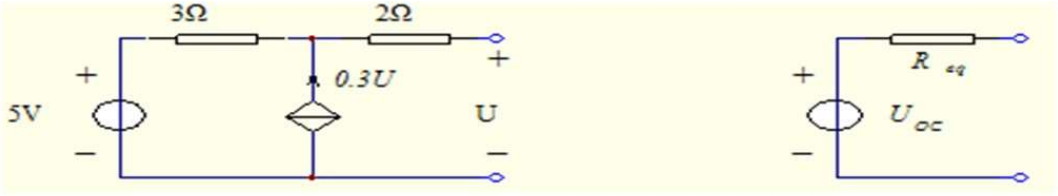
(2) 求 $u_C(t)$ $u_C(t) = u_C(0_+)e^{-\frac{t}{\tau}} = 4e^{-1.25t}V\ (t \geq 0)$

$i(t) = \frac{u_C(t)}{8} = 0.5e^{-1.25t}A\ (t \geq 0)$

(3) 求 $u(t)$ $u(t) = 5(i - 0.2i) = 4i = 2e^{-1.25t}V\ (t \geq 0)$

11-4 解: $u_c(0+) = u_c0-) = 0$

0+后等效电路如下

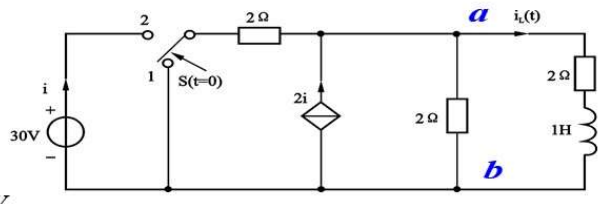


$$U_{OC} = 2 \times 0 + 3 \times 0.3U_{OC} + 5 \rightarrow U_{OC} = 50\ V$$

$$I_{SC} = \frac{5}{2+3} = 1\ A \qquad R_{eq} = \frac{U_{OC}}{I_{SC}} = 50\ \Omega \qquad \tau = R_{eq}C = 0.01s$$

$$u_C(t) = 50(1 - e^{-100t}) \qquad i(t) = C\frac{du_C}{dt} = e^{-100t}\ (A)$$

11-5 在图示电路中，开关S在t=0时由位置1倒向位置2，开关动作之前电路已处于稳态，求换路后电感电流 $i_L(t)$



解 (1) $i_L(0+) = i_L(0-) = 0\ A$

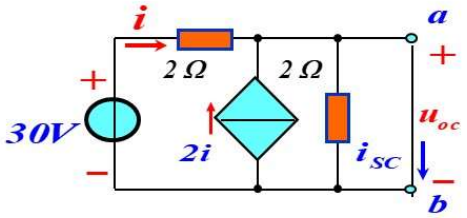
(2) 求戴维南等效电路

开路 $30 = 2i + 2 \cdot 3i$

$$i = \frac{30}{8} = 3.75\ A \qquad u_{oc} = 2 \cdot 3i = 22.5V$$

短路 $i = \frac{30}{2} = 15\ A \qquad i_{SC} = 3i = 45\ A$

$$R_{eq} = \frac{u_{OC}}{i_{SC}} = \frac{22.5}{45} = 0.5\Omega$$

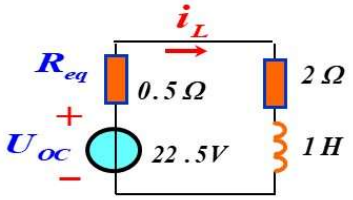


(3) 求时间常数 τ $\tau = \frac{L}{R_{eq}+2} = \frac{1}{2.5}s$

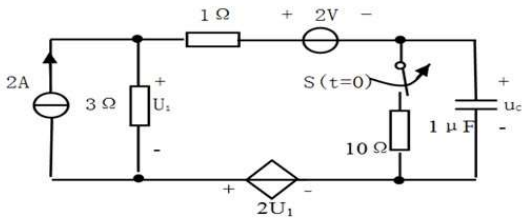
(4) 求 $i_L(\infty)$ $i_L(\infty) = \frac{u_{OC}}{2.5} = 9\ A$

(5) 求 $i_L(t)$

$$i_L(t) = i_L(\infty)(1 - e^{-\frac{t}{\tau}}) = 9(1 - e^{-2.5t})\ A(t \geq 0)$$



11-6 电路如图所示，开关在t<0时一直打开，在t=0时打开。求开关闭合后响应$uC(t)$。



解 (1) 求 $u(0_+)$

$$u(0_+)=2\mathrm{V}$$

(2) 求 $u(\infty)$

$$u_{oc}=16\mathrm{V},\quad R_{eq}=10\Omega \qquad u(\infty)=16\mathrm{V}$$

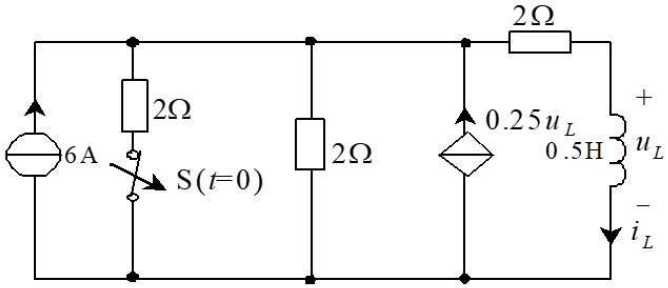
(3) 求时间常数 τ

$$\tau=10^{-5}\mathrm{s}$$

三要素为:

$$u(t)=u(\infty)+[u(0_+)-u_c(\infty)]e^{-\frac{t}{\tau}}$$
$$=16-14e^{-10^5 t}\mathrm{V}(t\geq 0)$$

11-7 电路如图所示，开关S在t=0时闭合，开关动作之前电路处于稳定状态，试求开关闭合后的$i_L(t)$。



解 (1) 求 $i_L(0_+)$

$$i_L(0_+)=2A$$

(2) 求 $i_L(\infty)$

$$u_{oc}=24\mathrm{V},\quad R_{eq}=8\Omega$$

$$i_L(\infty)=3\mathrm{A}$$

(3) 求时间常数 τ $\qquad \tau=\frac{1}{16}\mathrm{s}$

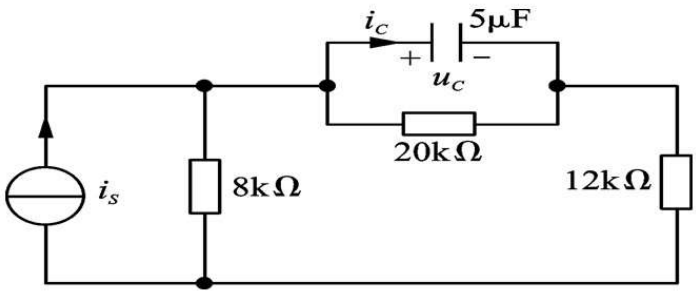
三要素为:

$$i_L(t)=i_L(\infty)+[i_L(0_+)-i_L(\infty)]e^{-\frac{t}{\tau}}$$
$$=3+(2-3)e^{-16t}\mathrm{A}(t\geq 0)$$

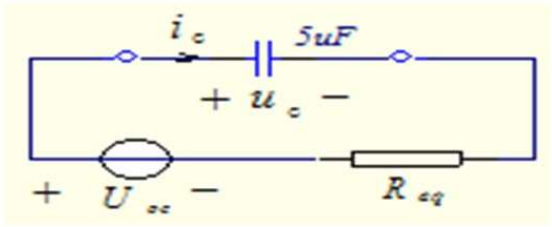
$$u_L=8e^{-16t}V\ (t\geq 0)$$

11-8 如图所示电路中电容原未充电， $i_S = \varepsilon(t)$ mA，

求响应 $u_C(t)$ 和 $i_C(t)$ 。



11-8 解: 等效电路如下 $u_C(0+) = u_C(0-) = 0$



$$R_{eq} = 20 \parallel (8+12) = 10k\Omega$$

$$U_{OC} = i_s \times \frac{8\times 32}{40} \times \frac{20}{20+12} = 4\,\delta(t)$$

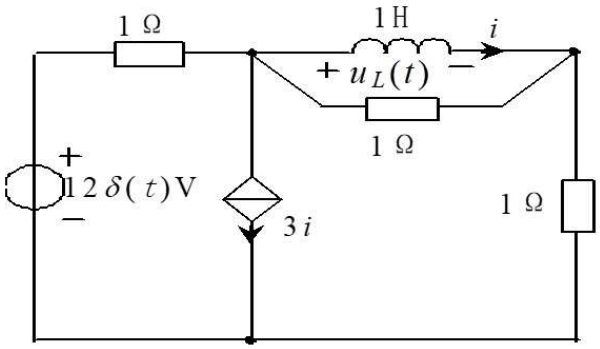
$$\tau = R_{eq}C = 0.05s$$

当 uoc=4 ε(t)时, 阶跃响应: $u_C(t) = 4(1-e^{-20t})\varepsilon(t)$

当 uoc=4 δ(t)时, 冲激响应: $u_C(t) = 80e^{-20t}\varepsilon(t)$

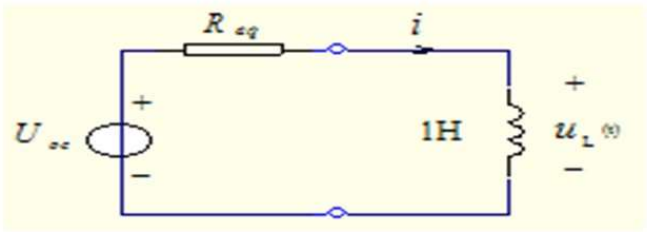
$$i(t) = C\frac{du_C}{dt} = -8e^{-20t}\,\varepsilon(t) + 0.4\delta(t)\,\text{mA}$$

11-9 求图所示$u_L(t)$的变化规律。



11-9 解: 等效电路如下

$$i(0+) = i(0-) = 0$$



$$U_{OC} = \frac{1}{3} \times 12\delta(t) = 4\,\delta\,(t)$$

$$I_{SC} = \frac{12}{5} \times \delta(t)$$

$$R_{eq} = \frac{U_{oc}}{I_{sc}} = \frac{5}{3}\Omega$$

当 uoc=4 ε(t)时, 阶跃响应:

$$i(t) = 2.4(1 - e^{-\frac{5}{3}t})\,\varepsilon\,(t)$$

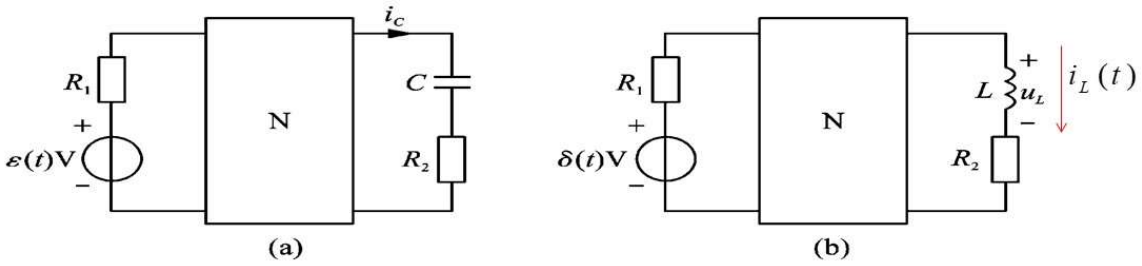
当 uoc=4 δ(t)时, 冲激响应:

$$i(t) = 4e^{-\frac{5}{3}t}\,\varepsilon\,(t)$$

$$u(t) = L\frac{di}{dt} = -\frac{20}{3}e^{-2t}\,\varepsilon\,(t) + 4\delta\,(t)$$

11-10 在图10-10(a)和(b)所示电路中，两个电阻网络N是相同的，已知 R1=R2=1kΩ, $C = 10\ \mu F$, L=4H, 图10-10(a)中的零状态响应

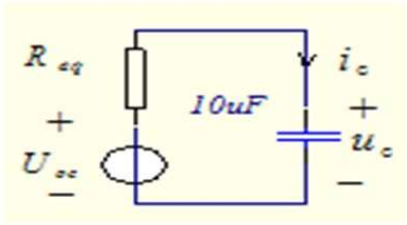
$i_C = \frac{1}{6}e^{-25t}$ mA，求图10-10(b)所示电路的零状态响应u_L



$$i_L(\infty) = i_C(0_+) = \frac{1}{6}mA$$

$$i_L(0_+) = i_C(\infty) = 0\,A$$

11-10 解:



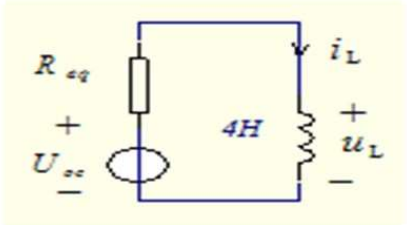
图(a)电路等效如下

$$u_C(t) = U_{oc}(1 - e^{-t/R_{eq}C})$$

$$i_c(t) = C\frac{du_C}{dt} = \frac{U_{oc}}{R_{eq}}e^{-t/R_{eq}C} = \frac{1}{6}e^{-25t}mA$$

$$\rightarrow \quad U_{oc} = \frac{2}{3}\varepsilon(t) \qquad R_{eq} = 4k\Omega$$

图(b)电源δ(t) →ε(t), 等效如左

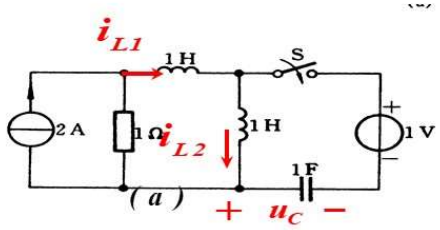


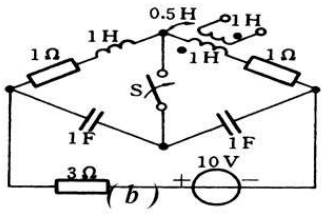
$$i_L(t) = \frac{1}{6}(1 - e^{-1000t})mA$$

图(b)电路中: $i_L(t) = \frac{1}{6}e^{-1000t}A$

$$u(t) = L\frac{di}{dt} = -\frac{2000}{3}e^{-1000t}\,\varepsilon(t) + \frac{2}{3}\delta(t)$$

12-1 图12-1（a）和（b）所示电路原已达稳定，t=0时把开关S合上，分别画出运算电路。





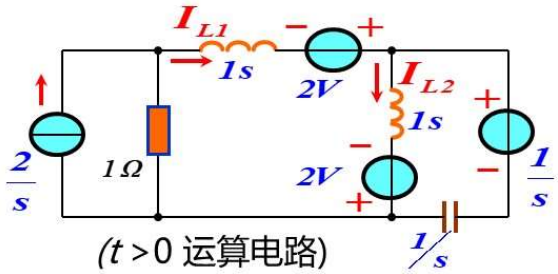
解 (a) (1) 求电容电压和电感电流初始值

$$i_{L1}(0_-) = i_{L2}(0_-) = 2\,A \qquad u_C(0_-) = 0V$$

(a) (2) 求运算电路模型

电感电流初始值附加电源: $Li_L(0_-)$

电容电压初始值附加电源: $\dfrac{U_C(0_-)}{s}$

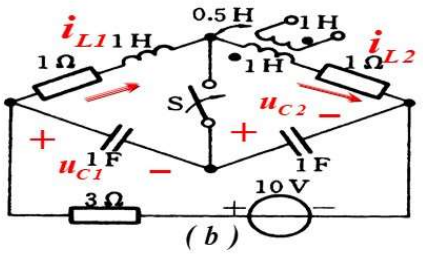


($t>0$ 运算电路)

解 (b) (1) 求电容电压和电感电流初始值

$$i_{L1}(0_-) = i_{L2}(0_-) = \frac{10}{3+2} = 2\,A$$

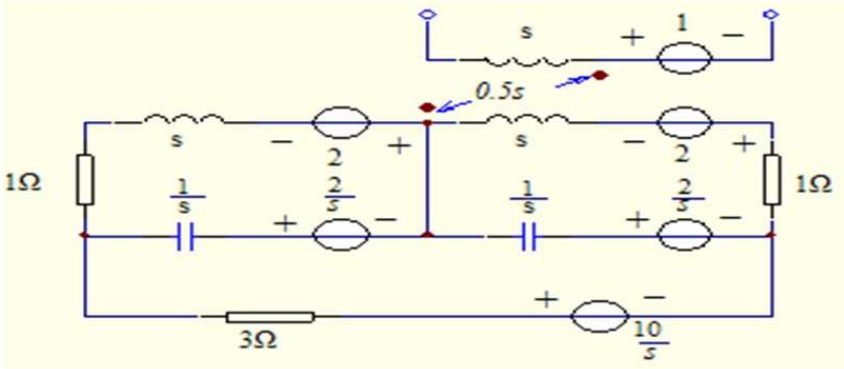
$$u_{C1}(0_-) = u_{C2}(0_-) = 2V$$



(b) (2) 求运算电路模型

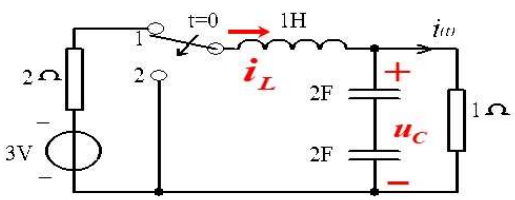
$$L_1 i_{L1}(0_-) = L_2 i_{L2}(0_-) = 2V$$

$$\frac{U_{C1}(0_-)}{s} = \frac{U_{C2}(0_-)}{s} = \frac{2}{s}$$



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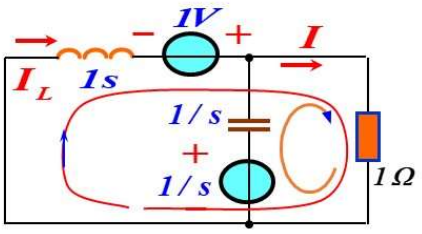
12-2 图示电路，开关S原在位置1很久且电路已达稳态，t=0时开关从位置1合至位置2，试用运算法求t>0之后，电流 $i(t)$



解 (1) 求电容电压和电感电流初始值

$$i_L(0_-)=\frac{3}{3}=1A \qquad u_C(0_-)=1V$$

(2) 画运算电路



(3) 回路法

$$\begin{cases} I_L(s)(1s+1)+1\cdot I(s)=1 \\ 1\cdot I_L(s)+(1+\frac{1}{s})I(s)=\frac{1}{s} \end{cases}$$

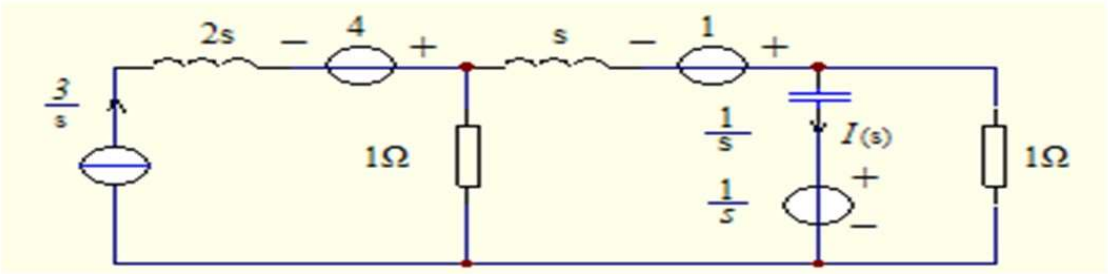
$$I(s)=\frac{s+1}{s^2+s+1}=\frac{s+\frac{1}{2}+\frac{1}{\sqrt{3}}\times\frac{\sqrt{3}}{2}}{(s+\frac{1}{2})^2+(\frac{\sqrt{3}}{2})^2}$$

(4) 反变换得i(t)

$$i(t)=e^{-0.5t}\cos\frac{\sqrt{3}}{2}t+\frac{1}{\sqrt{3}}e^{-0.5t}\sin\frac{\sqrt{3}}{2}t$$

12-3 解:

$$i_{L1}(0-)=2 \qquad i_{L2}(0-)=1 \qquad u_c(0-)=1$$



$$I(s)=\frac{1}{s^2+2s+2}=\frac{1}{(s+1)^2+(1)^2}$$

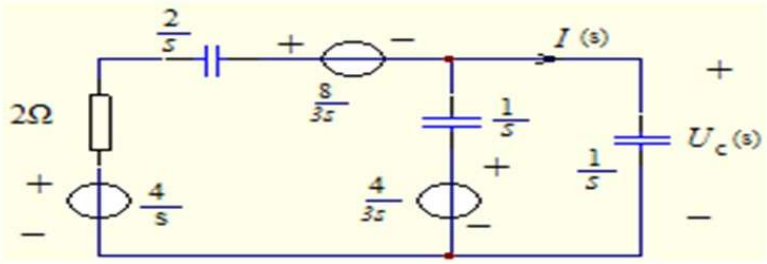
$$i(t)=e^{-t}\sin t \qquad t\geq 0$$

12-4 解:

$$u_{c1}(0-)=8/3$$

$$u_{c2}(0-)=4/3$$

$$u_{c3}(0-)=0$$



$$I(s)=\frac{8s+12}{12s+15} \qquad i(t)=\frac{2}{3}\delta(t)+\frac{1}{6}e^{-1.25t}$$

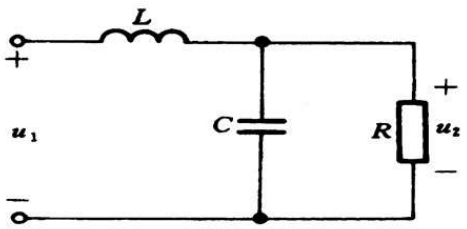
$$U(s)=\frac{8s+12}{s(12s+15)} \qquad u(t)=\frac{4}{5}\varepsilon(t)-\frac{2}{15}e^{-1.25t}$$

图12-6示电路为一低通滤波器，若已知冲激响应为

$$h(t)=\left[\sqrt{2}e^{-\frac{\sqrt{2}}{2}t}\sin\left(\frac{1}{\sqrt{2}}t\right)\right]\varepsilon(t)$$

求：（1）L、C值;

（2）幅频响应|H(jω)|-ω。



12-6 解:

$$H(s)=L[h(t)]=\frac{1}{s^2+\sqrt{2}s+1}$$

$$H(s)=\frac{R\parallel\frac{1}{sC}}{sL+R\parallel\frac{1}{SC}}=\frac{1}{LCS^2+\frac{L}{R}S+1}$$

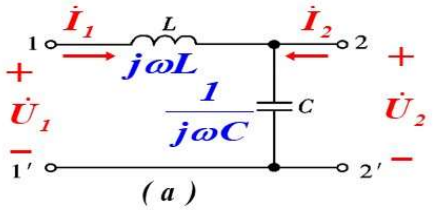
$$\Rightarrow \quad L=\sqrt{2}R \qquad C=\frac{\sqrt{2}}{2}R$$

$$H(j\omega)=\frac{1}{(j\omega)^2+\sqrt{2}(j\omega)+1}=\frac{1}{(1-\omega^2)+j\sqrt{2}\omega}$$

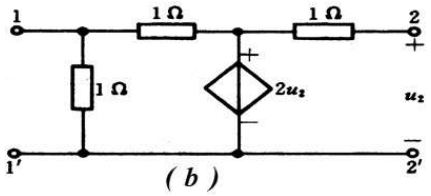
$$|H(j\omega)|=\frac{1}{\sqrt{(1-\omega^2)^2+(\sqrt{2}\omega)^2}}=\frac{1}{\sqrt{1+\omega^4}}$$

$$\theta=-\text{arctg}\frac{\sqrt{2}\omega}{1-\omega^2}$$

13-1 求图所示二端口的Y、Z 、T和H参数矩阵。



(a)



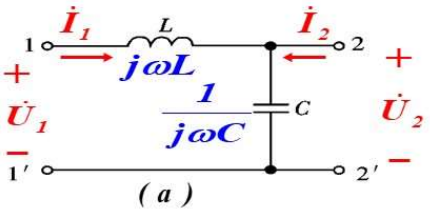
(b)

解 (a) (1) 求Y参数矩阵

$$\begin{cases}\dot{I}_1=\dfrac{\dot{U}_1-\dot{U}_2}{j\omega L}=\dfrac{1}{j\omega L}\dot{U}_1-\dfrac{1}{j\omega L}\dot{U}_2\\ \dot{I}_2=\dfrac{\dot{U}_2-\dot{U}_1}{j\omega L}+j\omega C\dot{U}_2=-\dfrac{1}{j\omega L}\dot{U}_1+\left(\dfrac{1}{j\omega L}+j\omega C\right)\dot{U}_2\end{cases}$$

$$\Rightarrow \quad Y=\begin{bmatrix}\dfrac{1}{j\omega L} & -\dfrac{1}{j\omega L}\\ -\dfrac{1}{j\omega L} & \left(\dfrac{1}{j\omega L}+j\omega c\right)\end{bmatrix}$$

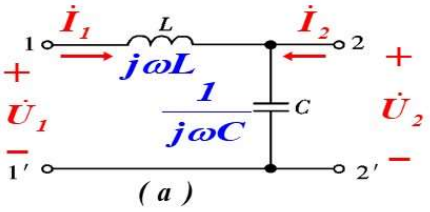
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(a) (2) 求Z参数矩阵

$$\begin{cases} \dot{U}_1 = j\omega L\dot{I}_1 + \dfrac{1}{j\omega C}(\dot{I}_1 + \dot{I}_2) = (j\omega L + \dfrac{1}{j\omega C})\dot{I}_1 + \dfrac{1}{j\omega C}\dot{I}_2 \\ \dot{U}_2 = \dfrac{1}{j\omega C}(\dot{I}_1 + \dot{I}_2) = \dfrac{1}{j\omega C}\dot{I}_1 + \dfrac{1}{j\omega C}\dot{I}_2 \end{cases}$$

$$\Rightarrow \quad Z = \begin{bmatrix} (j\omega L + \dfrac{1}{j\omega C}) & \dfrac{1}{j\omega C} \\ \dfrac{1}{j\omega C} & \dfrac{1}{j\omega C} \end{bmatrix}$$



(a) (3) 求T参数矩阵

$$\begin{cases} \dot{I}_1 = j\omega C\dot{U}_2 - \dot{I}_2 \\ \dot{U}_1 = j\omega L\dot{I}_1 + \dot{U}_2 = j\omega L(j\omega C\dot{U}_2 - \dot{I}_2) + \dot{U}_2 = (1 - \omega^2 LC)\dot{U}_2 - j\omega L\dot{I}_2 \end{cases}$$

$$\Rightarrow \quad T = \begin{bmatrix} (1 - \omega^2 LC) & j\omega L \\ j\omega C & 1 \end{bmatrix}$$

(a) (4) 求H参数矩阵

$$\begin{cases} \dot{U}_1 = j\omega L\dot{I}_1 + \dot{U}_2 \\ \dot{I}_2 = -\dot{I}_1 + j\omega C\dot{U}_2 \end{cases} \quad \Rightarrow \quad H = \begin{bmatrix} j\omega L & 1 \\ -1 & j\omega C \end{bmatrix}$$

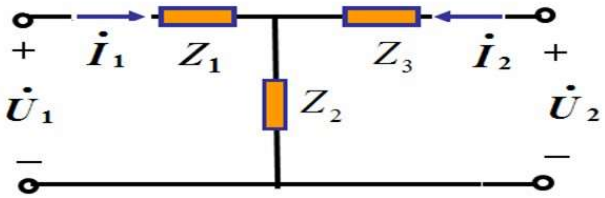
13-2 已知二端口的参数矩阵为

(1) $Z=\begin{bmatrix} 60/9 & 40/9 \\ 40/9 & 100/9 \end{bmatrix}\Omega$ (2) $Y=\begin{bmatrix} 5 & -2 \\ 0 & 3 \end{bmatrix}S$

试问该二端口是否有受控源，并求它的等效电路。

解 (1) $Z_{11}=\frac{60}{9}$ $Z_{22}=\frac{100}{9}$

$\because Z_{12}=Z_{21}=\frac{40}{9},\therefore$ 无受控源



$$Z'=\begin{bmatrix} Z_1+Z_2 & Z_2 \\ Z_2 & Z_2+Z_3 \end{bmatrix}==Z=\begin{bmatrix} 60/9 & 40/9 \\ 40/9 & 100/9 \end{bmatrix}$$

$$\Rightarrow Z_1=20/9\,\Omega, Z_2=40/9\,\Omega, Z_3=60/9\,\Omega$$

13-2 已知二端口的参数矩阵为

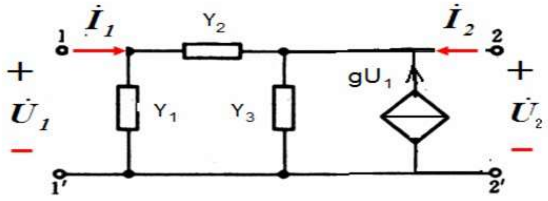
(1) $Z=\begin{bmatrix} 60/9 & 40/9 \\ 40/9 & 100/9 \end{bmatrix}\Omega$ (2) $Y=\begin{bmatrix} 5 & -2 \\ 0 & 3 \end{bmatrix}S$

试问该二端口是否有受控源，并求它的等效电路。

解 (2) $Y_{11}=5$ $Y_{22}=3$

$Y_{12}=-2$ $Y_{21}=0$

$\because Y_{12}\neq Y_{21},\therefore$ 有受控源

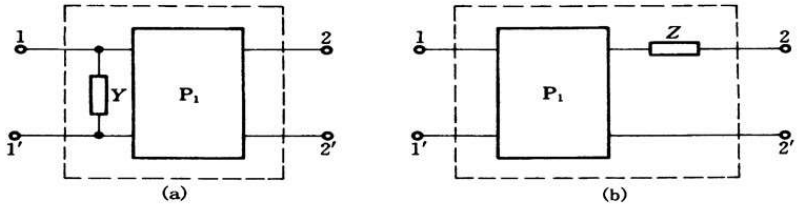


$$Y'=\begin{bmatrix} Y_1+Y_2 & -Y_2 \\ -Y_2-g & Y_2+Y_3 \end{bmatrix}==Y=\begin{bmatrix} 5 & -2 \\ 0 & 3 \end{bmatrix}$$

$$\Rightarrow Y_1=3S, Y_2=2S, Y_3=1S, g=-2S$$

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$$T_1 = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$



则(a):

$$[T] = [T_2][T_1] = \begin{bmatrix} 1 & 0 \\ Y & 1 \end{bmatrix}\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} C & B \\ YA+C & YB+D \end{bmatrix}$$

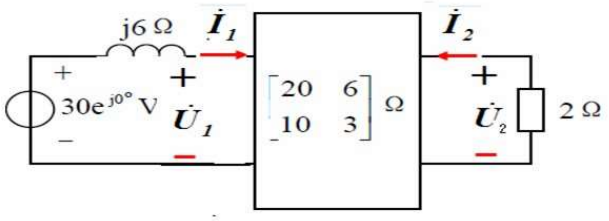
则(b):

$$[T] = [T_1][T_3] = \begin{bmatrix} A & B \\ C & D \end{bmatrix}\begin{bmatrix} 1 & Z \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} A & AZ+B \\ C & CZ+D \end{bmatrix}$$

13-4 正弦电路，求电压源发出的功率

解

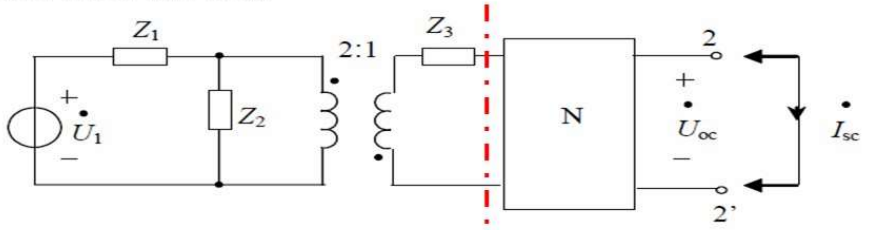
$$\begin{cases} \dot{U}_1 = 20\dot{I}_1 + 6\dot{I}_2 \\ \dot{U}_2 = 10\dot{I}_1 + 3\dot{I}_2 \\ \dot{U}_2 = -2\times\dot{I}_2 \\ j6\times\dot{I}_1 + \dot{U}_1 = 30\angle 0^\circ \end{cases}$$



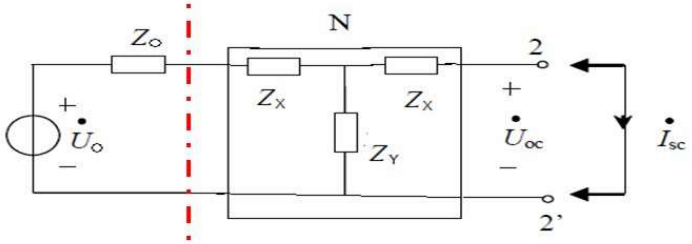
$$\Rightarrow \quad \dot{I}_1 = 3\angle -36.9^\circ$$

$$P = UI\cos\varphi = 30\times 3\times\cos 36.9^\circ = 54W$$

13-5: 试求网络N 的Z 参数



解 （1）如图等效



(2) $\dot{U}_o = 32\angle 180^\circ$

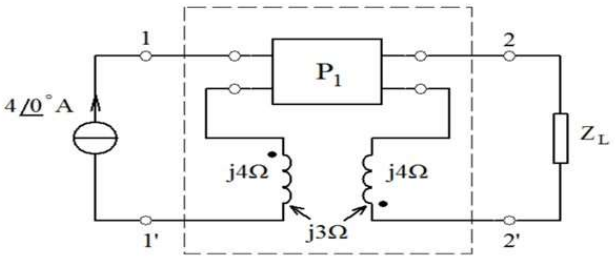
$Z_o = j2\Omega$

(3) $\begin{cases} \dot{U}_{OC} \\ \dot{I}_{SC} \end{cases} \Rightarrow Z_X = j2.8, Z_Y = j1.6$

(4) $Z = \begin{bmatrix} j4.4 & j1.6 \\ j1.6 & j4.4 \end{bmatrix}\Omega$

---

7 如图所示，虚框内为耦合互感及二端口网络 P_1 构成的复合二端口网络。已知二端口网络 P_1 的参数阵为 $Z = \begin{bmatrix} 4 & 3 \\ 3 & 4 \end{bmatrix}\Omega$。试求阻抗 Z_L 为何值时获得最大功率？并求此最大功率等于多少？



$$Z_1 = \begin{bmatrix} 4 & 3 \\ 3 & 4 \end{bmatrix}\Omega \quad Z_2 = \begin{bmatrix} j4 & -j3 \\ -j3 & j4 \end{bmatrix}\Omega \quad Z = \begin{bmatrix} 4+j4 & 3-j3 \\ 3-j3 & 4+j4 \end{bmatrix}\Omega$$

$$\dot{U}_1 = (4+j4)\dot{I}_1 + (3-j3)\dot{I}_2$$

$$\dot{U}_2 = (3-j3)\dot{I}_1 + (4+j4)\dot{I}_2$$

$$\dot{I}_1 = 4$$

$$Z_L = 4 - j4\Omega$$

$$P_{\max} = \frac{(12\sqrt{2})^2}{4*4} = 18\,\mathrm{W}$$